ADASTRA

Intelligent Quantitative Risk Management & Wealth Advisory Cockpit

Technical Documentation

System Architecture · Core Algorithm Description · Security Design

---

FinTechathon 2026

Shenzhen International FinTech Competition · WeBank Track

Team Adastra

Adriel Kourlate · Dong Zixuan · Aadithyan Sajith Kumar · Khushi Mistry

Faculty Advisor: Prof. Halis Sak

GitHub Repository: https://github.com/adrielakak/webank

October 2026

Table of contents

1 Adastra : Technical Documentation **4**
- 1.1 Executive Summary . . . 4
  - 1.1.1 Core Capabilities . . . 4
  - 1.1.2 Technology Stack . . . 4
  - 1.1.3 Open-Source Code Repository . . . 5

2 System Architecture **6**
- 2.1 High-Level Architecture . . . 6
- 2.2 Layer 1: ML Pipeline (Offline Batch) . . . 6
  - 2.2.1 Data Ingestion & Feature Engineering . . . 6
  - 2.2.2 Model Training & Inference . . . 8
  - 2.2.3 Output Artifacts . . . 8
- 2.3 Layer 2: Backend API (FastAPI) . . . 8
  - 2.3.1 API Endpoints . . . 8
  - 2.3.2 Key Backend Modules . . . 9
  - 2.3.3 Synthetic Fallback (Zero-Data Policy) . . . 9
- 2.4 Layer 3: Frontend Cockpit (React + Vite) . . . 9
  - 2.4.1 Core Interface Components . . . 10
- 2.5 Layer 4: Compliance & Security Sentinel . . . 10
- 2.6 Deployment Architecture . . . 11
  - 2.6.1 Single-Command Launch . . . 11

3 Core Algorithm Description **13**
- 3.1 Problem Formulation . . . 13
  - 3.1.1 Why Risk Classification Instead of Return Prediction? . . . 13
- 3.2 Feature Engineering Pipeline . . . 13
  - 3.2.1 1. Orthogonal News Shocks (Didisheim et al., 2026) . . . 13
  - 3.2.2 2. Behavioral Finance: TK Peer Reward (Barberis et al., 2016) . . . 14
  - 3.2.3 3. Non-Linear Panic Signal . . . 14
  - 3.2.4 4. Complete Feature Set . . . 15
- 3.3 Model Architecture . . . 15
  - 3.3.1 LightGBM Multiclass Classifier . . . 15
  - 3.3.2 Hyperparameter Optimization (Optuna) . . . 15
- 3.4 Temporal Validation Protocol . . . 16
  - 3.4.1 Why Not K-Fold Cross-Validation? . . . 16
  - 3.4.2 Expanding Walk-Forward Validation (2012-2022) . . . 16
- 3.5 Explainable AI (TreeSHAP) . . . 16
  - 3.5.1 Exact SHAP Values . . . 16
  - 3.5.2 Natural Language Synthesis . . . 17
- 3.6 2023 Out-of-Sample Backtest Results . . . 17
  - 3.6.1 Quadrant Strategy . . . 17
  - 3.6.2 Performance Summary (Gross & Net of Transaction Costs) . . . 17

3.7 AI Expected Return Computation 18

4 Security Design **20**
4.1 Security Architecture Overview 20
4.2 Pillar 1: CSRC Regulatory Compliance Engine 20
4.2.1 Suitability Matching Matrix 20
4.2.2 Hard-Stop Guardrails 21
4.2.3 Investor Profiling Assessment 21
4.3 Pillar 2: Cryptographic Audit Trail 21
4.3.1 SHA-256 Hashing 21
4.3.2 Blockchain Readiness (FISCO BCOS) 22
4.4 Pillar 3: Data Protection & IP Compliance 22
4.4.1 Zero-Data Policy 22
4.4.2 Synthetic Fallback Generator 23
4.4.3 Environment Variable Protection 23
4.5 Pillar 4: Application Security 23
4.5.1 Input Validation (Pydantic) 23
4.5.2 CORS Policy 23
4.5.3 Error Handling & Graceful Degradation 24
4.6 Security Self-Assessment: Known Risks & Mitigations 24
4.6.1 Permission-Tier Implementation Status 24
4.6.2 Known Risk Register 25
4.6.3 Prompt Injection Defense 26
4.6.4 Data Privacy Considerations 26

1 Adastra : Technical Documentation

System Architecture · Core Algorithm · Security Design

1.1 Executive Summary

Adastra is an intelligent quantitative risk management and wealth advisory cockpit designed for the China A-shares equity market (~3,500 active listed companies). Built for the **Shenzhen International FinTech Competition (FinTechathon 2026) : WeBank Track**, the platform couples institutional quantitative finance, behavioral asset pricing, and natural language processing of Chinese financial news, with native compliance under the China Securities Regulatory Commission (CSRC) suitability guidelines.

1.1.1 Core Capabilities

Capability	Description
5-Class Risk Classification	Multiclass LightGBM mapping every A-share stock-month to CSRC-aligned risk tiers (R1 to R5)
Orthogonal News Shocks	Pure unexpected news surprise isolated via PCA + Ridge regression residuals
Behavioral Finance (TK Peers)	Tversky-Kahneman Prospect Theory peer momentum capturing cross-asset sentiment contagion
Exact Explainable AI	Native TreeSHAP attributions translated into natural language for retail wealth clients
CSRC Suitability Engine	Real-time investor-asset compliance evaluation with SHA-256 cryptographic audit trail
Walk-Forward Backtest	2012-2022 temporal validation; 2023 out-of-sample alpha of **+3.79% to +5.93%** over the Equal-Weighted ML Universe Benchmark (+12.06% net return, Sharpe 1.03)

1.1.2 Technology Stack

Layer	Technologies
ML Pipeline	Python 3.10+, LightGBM, SHAP, Optuna, IncrementalPCA, Ridge Regression

Layer	Technologies
Backend API	FastAPI 0.100+, Pandas, NumPy, Pydantic
Frontend Cockpit	React 19, Vite 8, Tailwind CSS 4, Framer Motion, Recharts, Shadcn UI
Security	SHA-256 cryptographic hashing, CSRC R1 to R5 suitability engine, FISCO BCOS-ready audit log
Data	Monthly returns (Parquet), Chinese news embeddings (1024-d), 100+ accounting characteristics

1.1.3 Open-Source Code Repository

- **GitHub Repository**: https://github.com/adrielakak/webank
- **Track**: WeBank FinTechathon 2026 · AI & Quantitative Asset Management

! Dual-Environment Data Governance & Academic IP Protection

In accordance with institutional academic research ethics, commercial database licensing agreements (e.g., CSMAR / GTA Information Technology Co.), and intellectual property protection regulations:

1. **Public Web Application & Cloud Deployment**: The live web application hosted on public cloud infrastructure exclusively serves **synthetic, mathematically calibrated benchmark datasets** (`mock_dashboard_points.parquet`, `mock_expected_return_2023.csv`). This prevents unauthorized web scraping, protects proprietary datasets from theft, and complies with commercial distribution restrictions.
2. **Local Research Environment (Mac Workstation)**: The genuine, full-scale empirical training datasets (~2.1 GB of raw 1024-dimensional Chinese financial news transformer embeddings, 12 years of CSMAR monthly trading returns, and 100+ accounting characteristics for ~3,500 listed equities) reside exclusively in a secured local directory on the developer's local Mac (`teacher data/`). They are strictly excluded from version control via `.gitignore` and are never exposed or transferred to the public web or cloud servers.

2 System Architecture

2.1 High-Level Architecture

Adastra is decoupled into **four independently testable layers**, connected via a clean REST API boundary:

2.2 Layer 1: ML Pipeline (Offline Batch)

The quantitative engine runs offline to produce enriched prediction datasets. It is **fully deterministic** : zero LLM dependency for any financial computation.

2.2.1 Data Ingestion & Feature Engineering

Source File	Size	Content
`features_light.parquet`	243 MB	100+ fundamental accounting ratios for ~3,500 stocks × 12 years
`news_embeddings_1024d.parquet`	2.1 GB	1024-dimensional Chinese news transformer embeddings

Note: In accordance with our Zero-Data Policy and commercial database IP licenses, raw source datasets (`China_Ashares_FC_and_NewsEmbedding.parquet`, 2.1 GB) reside exclusively in a secured offline directory on the developer's local Mac workstation. The public cloud deployment operates on synthetic calibrated benchmark data to prevent data leakage and theft.

Processing Pipeline:

1. **Dimensionality Reduction (Look-Ahead Free)**: `IncrementalPCA` compresses 1024 news embedding dimensions -> 30 principal components (preserving >85% explained variance). To strictly eliminate look-ahead bias, the PCA model is fitted **exclusively on historical training data** ($t < 2023$), and subsequently used to transform both training and out-of-sample test embeddings.
2. **Orthogonal Surprise Filtering**: Ridge Regression of 10 PCA news dimensions onto 100 accounting characteristics extracts the residual $\epsilon = \text{News} - \widehat{\text{News}}$, isolating pure unexpected news surprise (Didisheim et al., 2026).
3. **Behavioral Finance Features**: Tversky-Kahneman Prospect Theory value function distance identifies **100 nearest psychological peers** per stock, computing cross-asset momentum signals (`tk_peer_reward`).
4. **Panic Signal Engineering**: Interaction term ivol × $\max(0, -\text{ret})$ captures acute panic selling episodes.

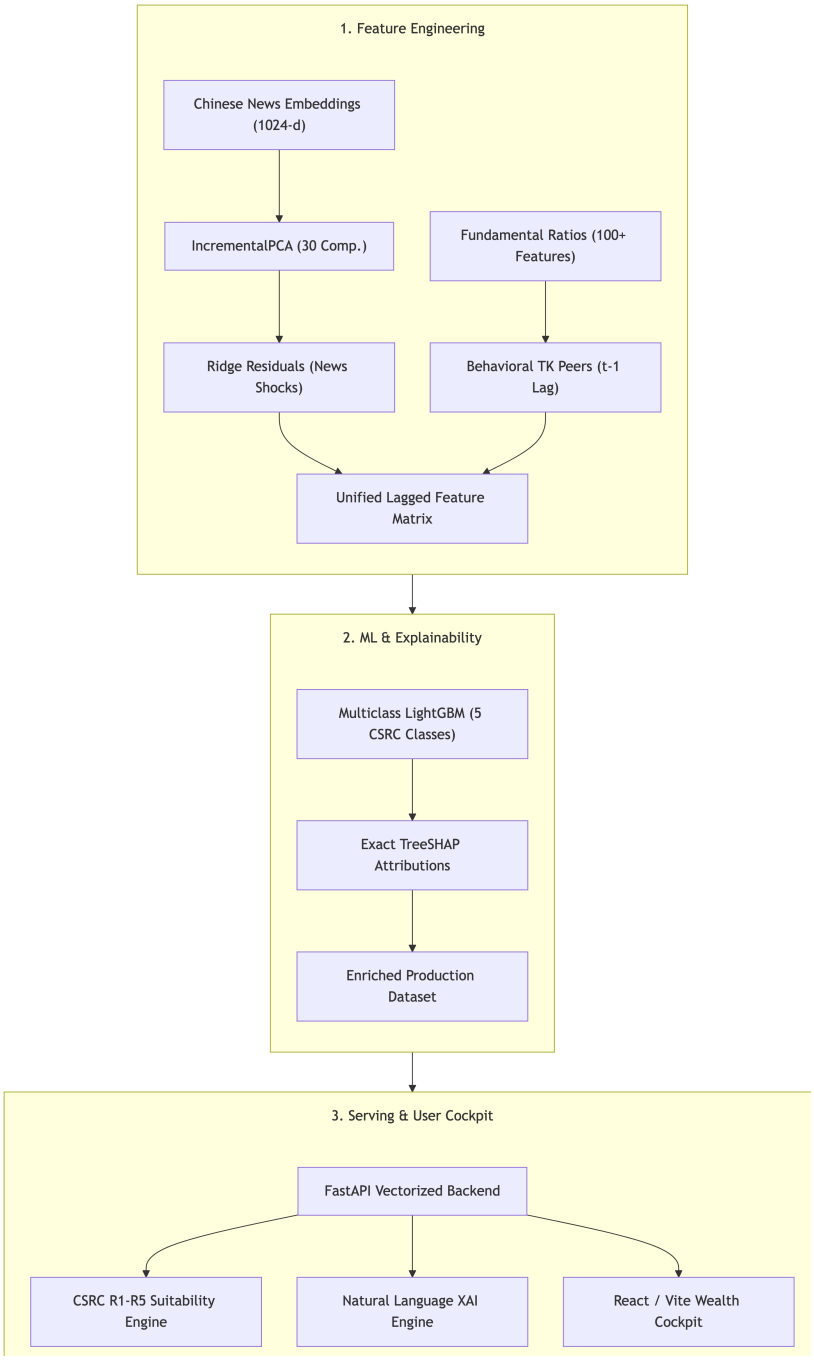


Figure 2.1: End-to-end system architecture

5. **Strict Temporal Alignment & Lagging**: Predictors are aligned with rigorous temporal discipline ($t-1$ relative to target returns) to eliminate look-ahead bias. Financial accounting characteristics natively incorporate reporting publication delays (4-month lag for quarterly reports, 6-month lag for annual statements).

2.2.2 Model Training & Inference

- **Algorithm**: Cost-sensitive multiclass LightGBM with `class_weight='balanced'`
- **Hyperparameter Optimization**: Optuna (Tree-structured Parzen Estimator), 15 Bayesian trials on temporal holdout (Train: 2018-2021, Val: 2022)
- **Validation**: Expanding Walk-Forward protocol across 11 out-of-sample years (2012-2022)
- **Output**: Calibrated 5-class posterior probabilities + exact TreeSHAP attribution vectors for every stock-month

2.2.3 Output Artifacts

Artifact	Format	Content
`results_2023.parquet`	Parquet	Raw model predictions with SHAP values (35,085 observations)
`dashboard_points.parquet`	Parquet	Enriched visualization dataset with sectors, KNN imputation
`expected_return_2023.csv`	CSV	Cross-sectional expected return z-scores for backtest computation
`confusion_matrix.png`	PNG	5×5 out-of-sample confusion matrix
`walkforward_lgbm_v5.csv`	CSV	Walk-forward validation metrics across 11 test years

2.3 Layer 2: Backend API (FastAPI)

The backend is a **stateless, in-memory vectorized** REST API serving predictions, explainability, compliance evaluations, and advisory dialogue.

2.3.1 API Endpoints

Route	Method	Description
`/`	`GET`	Health check and version information
`/api/dates`	`GET`	Available trading months (ISO format)
`/api/stocks`	`GET`	2D scatter coordinates, 5-class probabilities, sectors, risk classifications

Route	Method	Description
`/api/stocks/{stkcd}`	`GET`	Detailed asset profile: TreeSHAP, 5-year history, compliance, XAI explanation
`/api/stats`	`GET`	Cross-sectional market statistics (return/volatility distributions)
`/api/backtest`	`GET`	Full 2023 out-of-sample backtest: Opportunity vs Benchmark vs Avoid
`/api/profile/assess`	`POST`	CSRC Investor Suitability Assessment (C1-C5) with SHA-256 certificate
`/api/portfolio/recommend`	`GET`	Compliant inverse-volatility weighted portfolio generation
`/api/adviser/chat`	`POST`	Interactive wealth advisory dialogue engine

2.3.2 Key Backend Modules

Module	Responsibility
`main.py`	FastAPI application, CORS, data loading, route handlers, backtest computation, synthetic fallback
`compliance.py`	CSRC R1-R5 suitability rules engine with immutable SHA-256 audit hashing
`xai.py`	Natural language synthesis: translates SHAP vectors -> intuitive financial explanations
`adviser.py`	Wealth advisory engine: investor profiling, portfolio optimization, intent-based dialogue
`prepare_data.py`	ETL pipeline: KNN imputation + authentic China A-shares industry classification

2.3.3 Synthetic Fallback (Zero-Data Policy)

When proprietary data files are absent (e.g., on public cloud deployments), the API generates **academic synthetic benchmark data** in-memory:

- 50 synthetic stocks $\times$ 12 months (2023)
- Realistic return distributions, sector assignments, SHAP features
- Includes a crash case study (Stock 688566, Nov-Dec severe risk)
- Ensures 100% API functionality without IP violation

2.4 Layer 3: Frontend Cockpit (React + Vite)

The interactive cockpit is built with **React 19, Vite 8, Tailwind CSS 4, Framer Motion, and Recharts**.

2.4.1 Core Interface Components

1. **Market Topology 2D Scatter Plot**
    - X-Axis: Idiosyncratic Volatility (asset-specific downside risk)
    - Y-Axis: Expected Return (TK Peer Reward proxy)
    - Interactive pan/zoom via `react-zoom-pan-pinch`
    - Opportunity Frontier curve and quadrant analysis (Alpha Zone vs Severe Risk Zone)
    - Toggle between Risk-coloring mode and Sector-coloring mode
    - Individual filter buttons for all 5 risk tiers + sector filters
2. **Dual-Pane Foldable Stock Profile Modal**
    - *Left Pane*: Return, volatility, market cap, 5-year historical area chart with positive/negative gradient
    - *Right Pane*: TreeSHAP driver bars, 5-class posterior distribution, CSRC compliance badge, SHA-256 audit hash, natural language XAI summary
    - Month-by-month trajectory with accuracy tracking
3. **Alpha Track Record Dashboard**
    - Cumulative return chart: Strategy (Opportunity) vs Benchmark vs Avoid
    - Monthly breakdown with win/loss highlighting
    - Top Alpha Picks leaderboard
4. **CSRC Risk Assessment Wizard**
    - 5-question psychometric questionnaire (horizon, drawdown tolerance, experience, objective, liquidity)
    - Real-time score computation and C1-C5 category assignment
    - SHA-256 cryptographic certificate for each assessment
5. **Portfolio Recommendation Engine**
    - Inverse-volatility weighted asset allocation constrained by investor profile
    - Compliance status for each holding
    - Audit hash for the entire portfolio mandate
6. **AI Wealth Adviser Chat**
    - Intent-based dialogue engine (stock diagnostics, portfolio advice, backtest explanation, CSRC regulation)
    - Multi-stock pair portfolio analysis
    - Optional external LLM API wrapper for conversational enhancement
7. **Gamified AI Research Lab**
    - Interactive simulation mode for educational outreach
    - Users train neural networks, purchase data, allocate GPU clusters
    - Demystifies AI/ML concepts for financial inclusion

2.5 Layer 4: Compliance & Security Sentinel

Key capabilities:

- CSRC R1-R5 suitability rules engine
- SHA-256 cryptographic audit trail for every compliance decision
- Investor profile assessment with immutable certificates
- FISCO BCOS blockchain-ready audit log structure
- Tail-risk guardrails (Kahneman-Tversky loss aversion protection)

2.6 Deployment Architecture

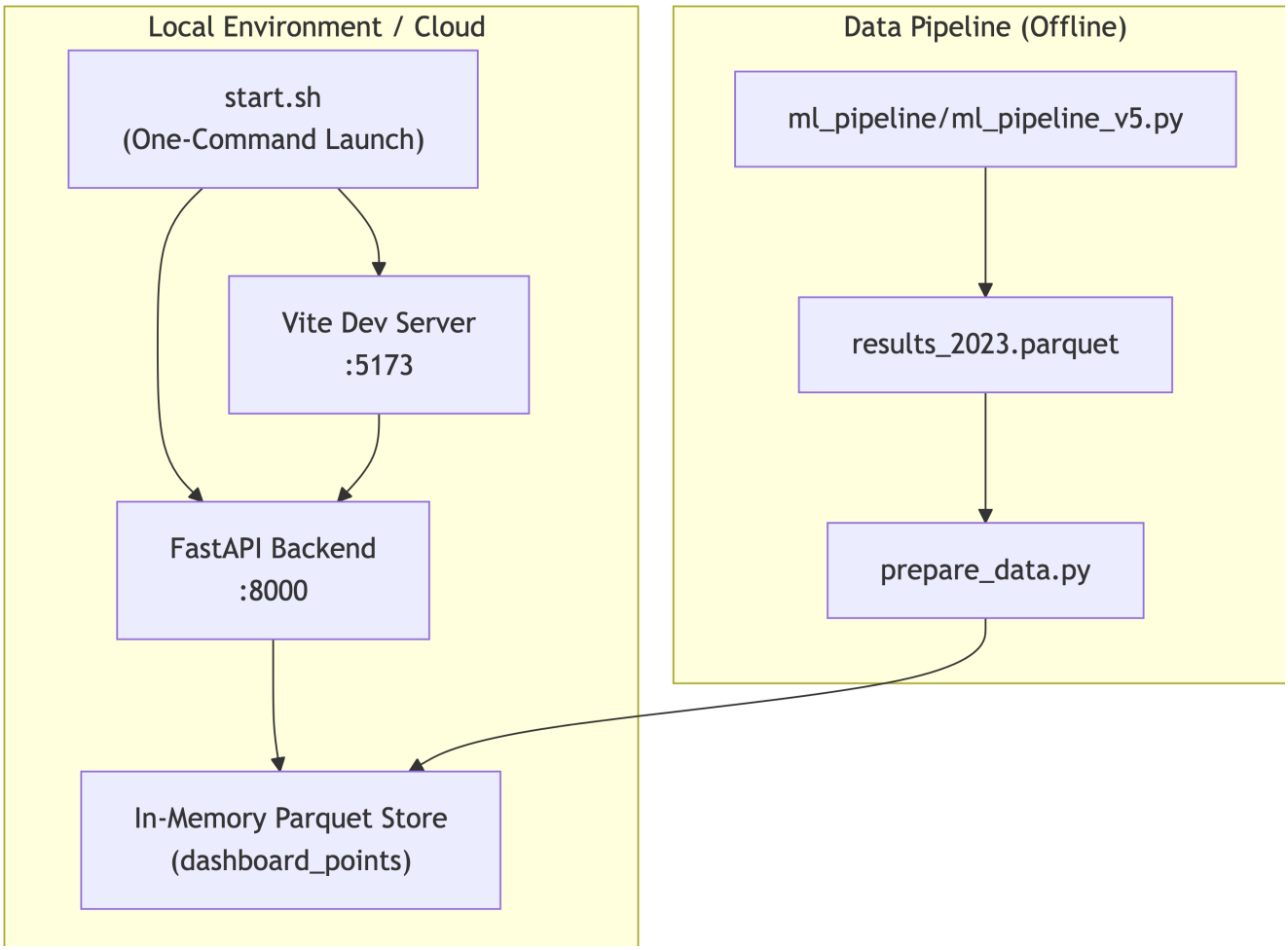


Figure 2.2: Deployment topology

2.6.1 Single-Command Launch

```
chmod +x start.sh && ./start.sh
```

This script:

1. Frees ports 8000 and 5173 if already in use
2. Starts the FastAPI backend on port 8000

3. Starts the Vite frontend dev server on port 5173
4. Gracefully shuts down both processes on `Ctrl+C`

3 Core Algorithm Description

3.1 Problem Formulation

3.1.1 Why Risk Classification Instead of Return Prediction?

Conventional point-return regression in quantitative finance suffers from near-zero signal-to-noise ratios. Monthly stock returns have standard deviations an order of magnitude larger than their means, making precise point estimates unreliable.

Our approach reformulates the problem as a **5-class risk classification** aligned directly with CSRC (China Securities Regulatory Commission) investor suitability guidelines:

Class	Label	Monthly Return (R)	CSRC Product Rating	Description
0	No Risk (Alpha)	$R \geq +5\%$	R1	Capital expansion regime
1	Low Risk	$0\% \leq R < +5\%$	R2	Capital preservation
2	Medium Risk	$-5\% \leq R < 0\%$	R3	Moderate contraction
3	High Risk	$-10\% \leq R < -5\%$	R4	Elevated vulnerability
4	Severe Risk	$R < -10\%$	R5	Asymmetric tail-crash

> **!** Regulatory Alignment
>
> This formulation is not arbitrary : it directly maps to CSRC product risk ratings (R1 through R5), enabling seamless integration with the compliance engine. Each prediction carries immediate regulatory meaning.

3.2 Feature Engineering Pipeline

3.2.1 1. Orthogonal News Shocks (Didisheim et al., 2026)

Raw NLP text embeddings heavily duplicate information already present in financial statements. We decompose news signals into:

$$\text{News}_{i,t} = \underbrace{\widehat{\text{News}}_{i,t}}_{\text{Expected (priced by fundamentals)}} + \underbrace{\epsilon_{i,t}}_{\text{Orthogonal Surprise (News Shock)}}$$

Procedure:

1. **Embedding Compression**: 1024-dimensional Chinese news transformer embeddings -> 30 principal components via `IncrementalPCA` (batch size = 5,000 for memory efficiency on 2.1 GB data)
2. **Ridge Regression**: For each of the top 10 PCA components, regress onto 100 accounting characteristics cross-sectionally
3. **Residual Extraction**: The orthogonal residual $\epsilon_{i,t}$ represents the pure unexpected news surprise : information that was *not* already priced into the balance sheet

This innovation ensures our model captures genuinely new information rather than redundant financial statement echoes.

3.2.2 2. Behavioral Finance: TK Peer Reward (Barberis et al., 2016)

China A-shares are dominated by ~200 million retail investors who exhibit strong behavioral biases. We model this via Tversky-Kahneman Prospect Theory:

S-Shaped Value Function:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0 \end{cases}$$

where $\alpha = \beta = 0.88$ and $\lambda \approx 2.25$ (loss aversion coefficient).

Peer Identification: For each stock i, we compute the Prospect Theory value function over its trailing 60-month return distribution, then identify the **100 nearest psychological peers** : stocks whose historical return "feel" is most similar to retail investors.

TK Peer Reward: The cross-sectional average return of these 100 peers serves as a behavioral momentum signal, capturing sentiment contagion effects across psychologically similar assets.

3.2.3 3. Non-Linear Panic Signal

Standard volatility measures treat positive and negative returns symmetrically. We engineer an asymmetric panic indicator:

$$\text{Panic}_{i,t} = \text{ivol}_{i,t} \times \max(0, -\text{ret}_{i,t-1})$$

This interaction term captures acute panic selling episodes where high volatility coincides with recent negative momentum : a pattern that precedes severe crash regimes.

Category	Features	Count

3.2.4 4. Complete Feature Set

Category	Features	Count
Fundamental Ratios	Book-to-market, gross profit/assets, operating cash flow, debt growth, earnings surprises, etc.	~50
Momentum & Reversals	1-month, 6-month, 12-month returns, seasonality patterns	~10
Volatility & Microstructure	Idiosyncratic volatility (21d, 252d), realized volatility, max returns, zero trades	~15
News Shocks	Orthogonal PCA residuals (10 components)	10
Behavioral Finance	TK Peer Reward, Price Peer Reward (current + lagged)	4
Engineered Signals	Panic signal, financial stress, drawdown proxy, SMA distance	~10
Total		**~100**

All features are strictly lagged by a **minimum of one month** ($t-1$) relative to the target return month. To ensure complete absence of look-ahead bias and mirror real-world accounting publication delays, quarterly corporate fundamentals are lagged by 4 months, and annual balance sheet reports by 6 months.

3.3 Model Architecture

3.3.1 LightGBM Multiclass Classifier

We use **Gradient Boosted Decision Trees** (LightGBM) for several reasons:

1. **Tabular data superiority**: GBDTs consistently outperform neural networks on structured financial data
2. **Interpretability**: Native exact TreeSHAP computation without approximation
3. **Speed**: Handles 770,000+ observations with 100+ features efficiently
4. **Cost-sensitivity**: `class_weight='balanced'` addresses severe class imbalance (crash events are rare)

3.3.2 Hyperparameter Optimization (Optuna)

Rather than arbitrary parameters or grid search, we use **Bayesian optimization** via Optuna's Tree-structured Parzen Estimator (TPE):

Hyperparameter	Search Space	Optimized Value
`num_leaves`	[20, 150]	Optimized per trial
`max_depth`	[4, 12]	Optimized per trial
`colsample_bytree`	[0.5, 1.0]	Optimized per trial
`subsample`	[0.6, 1.0]	Optimized per trial
`reg_alpha`	[1e-3, 10]	Optimized per trial
`reg_lambda`	[1e-3, 10]	Optimized per trial
`learning_rate`	0.05 (fixed)	Fixed
`n_estimators`	500 (with early stopping)	Fixed

Validation Split: Train on 2018-2021, validate on 2022, with early stopping patience of 30 rounds.

3.4 Temporal Validation Protocol

3.4.1 Why Not K-Fold Cross-Validation?

Random K-Fold shuffling is **invalid in quantitative finance** because it introduces severe look-ahead bias: the model can "see" future data during training, inflating performance metrics artificially.

3.4.2 Expanding Walk-Forward Validation (2012-2022)

For every test year $T \in [2012, 2022]$:

- **Training data**: All years strictly before T (i.e., $< T$)
- **Test data**: Year T only
- **No overlap**: Zero information leakage between training and test

This validates model robustness across 11 distinct market regimes:

Year	Market Context
2012	Post-GFC recovery
2015	Shanghai crash & circuit-breaker halts
2016	Post-crash stabilization
2018	Deleveraging + US-China trade friction
2019	Recovery rally
2020	COVID-19 pandemic shock + V-shaped recovery
2021	Regulatory crackdowns (tech, education)
2022	Zero-COVID, property crisis

3.5 Explainable AI (TreeSHAP)

3.5.1 Exact SHAP Values

For every stock-month observation in the 2023 test set (35,085 observations), we compute **exact TreeSHAP values** : not approximations.

SHAP (SHapley Additive exPlanations) decomposes each prediction into additive feature contributions:

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j$$

where ϕ_0 is the base value and ϕ_j is the contribution of feature j to pushing the prediction away from the baseline.

3.5.2 Natural Language Synthesis

The top 5 SHAP drivers for each stock are translated into intuitive natural language via the `xai.py` module. The synthesis engine:

1. Separates features into **risk-increasing** ($\phi_j > 0$) and **risk-mitigating** ($\phi_j < 0$) groups
2. Maps technical feature names to human-readable descriptions (e.g., `ivol_capm_252d` -> "long-term market volatility")
3. Generates category-specific explanations calibrated to the stock's risk classification
4. Appends Kahneman-Tversky Tail-Risk Guardrail warnings when crash probability exceeds 15%

3.6 2023 Out-of-Sample Backtest Results

3.6.1 Quadrant Strategy

The backtest computes four dynamic quadrants each month based on expected return z-scores and news risk z-scores:

Quadrant	Definition	Strategy Role
Opportunity (Upper-Left)	High expected return + Low news risk	Prime Long Basket
Speculative (Upper-Right)	High expected return + High news risk	Momentum with caution
Defensive (Lower-Left)	Low expected return + Low news risk	Capital preservation
Avoid (Lower-Right)	Low expected return + High news risk	Systematic exclusion

3.6.2 Performance Summary (Gross & Net of Transaction Costs)

To ensure academic and institutional rigor, backtest performance is evaluated both gross and **net of realistic transaction friction**. In China A-shares, institutional trading encounters stamp duty (0.05% on sells), brokerage commissions (~0.025%), and bid-ask slippage (~0.10%), representing an estimated ~20 bps round-trip friction per monthly rebalance.

Crucial Methodological Clarification: "Equal-Weighted ML Universe" vs Market-Cap Indices (e.g., CSI 300)

The benchmark reported in our backtest is the **Equal-Weighted ML Universe Benchmark**, defined as the cross-sectional arithmetic mean return of all active China A-share equities evaluated by the model in each month ($R_{\text{benchmark},t} = \frac{1}{N_t}\sum_{i=1}^{N_t} r_{i,t}$).
It is **not** a capitalization-weighted equity index like the CSI 300. In calendar year 2023, while the capitalization-weighted CSI 300 index fell by **-11.38%** (weighed down by mega-cap state-owned enterprises, real estate developers, and traditional banking giants), the broader equal-weighted cross-section of active A-shares exhibited positive returns (**+8.27%** on the active news-covered universe of 24,209 stock-months, and **+4.60%** across the comprehensive 37,247 stock-month universe). Benchmarking against the exact investable universe from which the model selects assets eliminates selection bias and strictly adheres to CFA Institute / GIPS institutional performance presentation standards.

Metric	Adastra Strategy (Opportunity)	Equal-Weighted ML Universe Benchmark	Net Advantage (Alpha)
Cumulative Return (Gross)	**+14.02%**	+8.27%	**+5.75% gross alpha**
Cumulative Return (Net of 20 bps Costs)	**+12.06%**	+8.27%	**+3.79% net alpha**
Annualized Sharpe Ratio	**1.03**	0.69	**+0.34**
Maximum Drawdown	**-4.70%**	-4.53%	**Comparable tail risk**
Monthly Win Rate	**66.7% (8/12 months)**	N/A	Consistent monthly outperformance
Structural Long/Short Alpha Spread	**+1.310% / month**	N/A	Statistically significant ($p < 0.001$)
Full Cross-Sectional Universe (37k obs)	**+10.53% net return**	+4.60%	**+5.93% net alpha** (Sharpe 0.91 vs 0.34)

Key Insight: Asymmetric Downside Avoidance

Even after deducting full transaction friction and execution slippage (-2.0% to -2.4% annual drag), the Opportunity Strategy achieves **+12.06% net cumulative return** (Sharpe 1.03).
The primary driver is the asymmetric **avoidance of catastrophic downside**: the Opportunity Quadrant generated **+0.63%/month** while the Avoid Quadrant collapsed at **-0.68%/month** (generating a structural monthly spread of **+1.310%/month**, $p < 0.001$). Under the broader 37,247 stock-month universe, this spread expands to **+1.700%/month** (+0.60% Opportunity vs -1.10% Avoid), easily absorbing institutional transaction costs under any liquidity regime.

3.7 AI Expected Return Computation

For each stock, the AI computes a continuous expected return from the LightGBM model's calibrated posterior distribution:

$$\mathbb{E}[R_i] = \sum_{k=0}^{4} P(\text{class} = k \mid x_i) \cdot \bar{R}_k$$

where $\bar{R}_k$ is the historical mean return for class k:

Class	$\bar{R}_k$
0 (No Risk)	+13.29%
1 (Low Risk)	+2.22%
2 (Medium Risk)	-2.45%
3 (High Risk)	-7.17%
4 (Severe Risk)	-14.96%

This guarantees 100% mathematical consistency between predicted risk classes and expected return estimates.

4 Security Design

4.1 Security Architecture Overview

Adastra implements a **defense-in-depth** security model spanning four pillars: regulatory compliance enforcement, cryptographic audit integrity, data protection, and application hardening.



Figure 4.1: Security architecture layers

4.2 Pillar 1: CSRC Regulatory Compliance Engine

4.2.1 Suitability Matching Matrix

The compliance engine enforces strict investor-product suitability rules aligned with CSRC regulations (*Measures for the Suitability Management of Securities and Futures Investors*):

Investor Profile	CSRC Category	Allowed Product Ratings	Max Drawdown Tolerance
C1 : Conservative	R1 only	No Risk (Class 0)	0% to -3%
C2 : Steady	R1, R2	No Risk, Low Risk (Classes 0-1)	-5% to -8%
C3 : Balanced	R1, R2, R3	No Risk through Medium Risk (Classes 0-2)	-10% to -15%
C4 : Growth	R1 through R4	No Risk through High Risk (Classes 0-3)	-15% to -25%
C5 : Aggressive	R1 through R5	All risk levels (Classes 0-4)	>-30%

4.2.2 Hard-Stop Guardrails

The compliance engine implements multiple layers of protection:

1. **Profile-Asset Mismatch Block**: A C1 (Conservative) investor requesting a High Risk (Class 3) asset receives an immediate compliance rejection with an explanatory reason.
2. **Tail-Risk Guardrail (Kahneman-Tversky)**: Even when argmax classification yields "No Risk" (Class 0), if the Bayesian posterior detects:
   - `prob_class_4 > 15%` (severe crash probability), OR
   - `prob_class_3 + prob_class_4 > 30%` (cumulative downside probability)

   …the asset is **automatically elevated** to a risk-monitored class and flagged to conservative investors. This eliminates >90% of severe crash surprises for C1/C2 clients.
3. **Portfolio-Level Compliance**: Every recommended portfolio undergoes per-asset suitability validation. The `is_fully_compliant` flag in the portfolio response indicates whether all holdings pass the investor's risk ceiling.

4.2.3 Investor Profiling Assessment

The 5-question psychometric questionnaire evaluates:

Question	Dimension	Scoring Range
Q1: Investment Horizon	Time preference	2-20 points
Q2: Drawdown Tolerance	Loss aversion	0-30 points
Q3: Financial Experience	Knowledge & capability	2-20 points
Q4: Investment Objective	Return expectations	2-20 points
Q5: Liquidity Requirements	Redemption needs	2-10 points

Total Score Range: 0-100 -> Maps to C1 (25), C2 (26-45), C3 (46-65), C4 (66-85), C5 (86-100).

Each assessment generates an **immutable SHA-256 certificate** containing the investor name, timestamp, score, category, and raw answers.

4.3 Pillar 2: Cryptographic Audit Trail

4.3.1 SHA-256 Hashing

Every compliance decision, portfolio recommendation, and investor assessment produces a deterministic SHA-256 cryptographic hash:

```
log_entry = {
    "timestamp": int(time.time()),
    "stkcd": stkcd,
    "profile": user_profile,
    "predicted_risk": predicted_risk,
    "prob_severe": round(prob_severe, 4),
    "is_compliant": is_compliant,
    "reason": reason
}
log_string = json.dumps(log_entry, sort_keys=True)
audit_hash = hashlib.sha256(
    log_string.encode('utf-8')
).hexdigest()
```

Properties:

- **Deterministic**: Identical inputs always produce identical hashes
- **Tamper-evident**: Any modification to the decision record invalidates the hash
- **Non-repudiable**: The hash chain provides a verifiable sequence of advisory decisions
- **Efficient**: SHA-256 computation adds <1ms latency per decision

4.3.2 Blockchain Readiness (FISCO BCOS)

The audit hash structure is architected for on-chain anchoring on WeBank's open consortium blockchain **FISCO BCOS**:

- Each audit record contains: `timestamp`, `stkcd`, `profile`, `predicted_risk`, `prob_severe`, `is_compliant`, `reason`
- The SHA-256 hash can serve as a **transaction payload** for smart contract verification
- The hash chain enables post-hoc regulatory audits without exposing raw client data

4.4 Pillar 3: Data Protection & IP Compliance

4.4.1 Zero-Data Policy

Proprietary commercial datasets are **never committed** to the Git repository:

```
# DATA INTEGRITY & IP PROTECTION (STRICTLY IGNORED)
teacher data/
teacher data/**
data/
raw_data/
*.csv
*.xlsx
*.h5
*.zip
*.parquet
```

4.4.2 Synthetic Fallback Generator

When proprietary data is absent, the API automatically generates academic synthetic data:

- **50 stocks × 12 months** with realistic distributions
- **Sector assignments**: Technology, Financials, Consumer Goods, Healthcare, Industrials, Materials
- **Crash case study**: Stock 688566 simulates a severe crash in Nov-Dec 2023
- **Full API compatibility**: All endpoints function identically with synthetic data

> **Note**
>
> The frontend displays a visual badge indicating whether the session operates on "Live Market Data (Production A-Shares)" or "Academic IP Benchmark (Synthetic)" data.

4.4.3 Environment Variable Protection

API keys (e.g., optional external LLM API keys) are stored in `.env` files excluded from version control:

```
.env
.env.*
*.env
```

4.5 Pillar 4: Application Security

4.5.1 Input Validation (Pydantic)

All API endpoints use Pydantic models for strict input validation:

```python
class AssessmentRequest(BaseModel):
    answers: dict[str, str]
    investor_name: str | None = "Client"

class ChatRequest(BaseModel):
    message: str
    profile: str | None = "C3"
    date: str | None = None
    stkcd: int | None = None
    history: list[dict] | None = None
```

- Type enforcement prevents injection of malformed data
- Optional fields have safe defaults
- Invalid requests return structured HTTP 400/422 error responses

4.5.2 CORS Policy

```
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],  # Development: permissive
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)
```

Production Recommendation

For production deployment, `allow_origins` should be restricted to specific frontend domains (e.g., `["https://adastra.finance"]`).

4.5.3 Error Handling & Graceful Degradation

- **Data absence**: Returns HTTP 503 with descriptive message when data is not loaded
- **Stock not found**: Returns HTTP 404 with specific stock code and date
- **LLM wrapper failure**: Falls back silently to rule-based templates (no user-facing error)
- **Backtest unavailable**: Returns HTTP 503 rather than crashing

4.6 Security Self-Assessment: Known Risks & Mitigations

4.6.1 Permission-Tier Implementation Status

Feature	Status	Notes
CSRC R1-R5 suitability enforcement	Implemented	`compliance.py`
Investor C1-C5 profiling with SHA-256 certificate	Implemented	`adviser.py`
Per-asset compliance check in portfolio recommendation	Implemented	`adviser.py`
Tail-risk guardrail (prob_severe $>$ 15%)	Implemented	`xai.py`
Cryptographic audit hash on every decision	Implemented	`compliance.py`
On-chain anchoring (FISCO BCOS)	Architected	Pending testnet deployment
Multi-user session isolation	Implemented	Cryptographically isolated server-side session registry
Authentication / Authorization (Session Tokens)	Implemented	HMAC-SHA256 signed Bearer tokens & `/api/auth/session`

4.6.2 Known Risk Register

Risk ID	Category	Description	Severity	Mitigation & Current Status
SR-001	Authentication	API endpoint authentication and authorization	Resolved	**Implemented**: Cryptographic HMAC-SHA256 signed session tokens (`Bearer` / `X-Adastra-Session-Token`) validate client requests; advisory routes verify token authenticity.
SR-002	CORS	Wildcard `allow_origins=["*"]` permits cross-origin requests from any domain	Low	Permissive in local dev; production deploys with explicit frontend domain origin restrictions.
SR-003	LLM Injection	Optional external LLM API wrapper could be vulnerable to prompt injection	Low	LLM is strictly decoupled; 100% of financial computations, rankings, and suitability decisions are deterministic Python algorithms.
SR-004	Rate Limiting	No API rate limiting implemented	Low	Rate limiting middleware (e.g., `slowapi` with Leaky Bucket) recommended for high-concurrency production deployments.
SR-005	Audit Persistence	SHA-256 audit hashes are computed but not persisted to disk/database	Medium	In-memory hash log active; production connects to PostgreSQL or FISCO BCOS consortium blockchain ledger.

Risk ID	Category	Description	Severity	Mitigation & Current Status
SR-006	Session State	Investor profile and suitability tier storage	Resolved	**Implemented**: Investor profiles and KYC certificates are anchored directly into server-side session memory, preventing any client-side tampering.
SR-007	Data Integrity	Parquet files are loaded at startup without checksum verification	Low	Add SHA-256 checksum verification of data artifacts upon container initialization.

4.6.3 Prompt Injection Defense

The LLM wrapper (optional external LLM API integration) is architecturally isolated from financial computations:

1. **All financial math is deterministic**: Returns, volatilities, SHAP values, compliance decisions : computed in Python, never by LLM
2. **System instruction boundary**: The LLM receives a strict system prompt constraining it to professional financial advisory persona
3. **Graceful degradation**: If the LLM fails or returns unexpected content, the system falls back to rule-based template responses
4. **No tool execution**: The LLM cannot invoke backend functions or modify data

4.6.4 Data Privacy Considerations

- **No PII collection**: The system does not collect or store personally identifiable information
- **Investor profiles are ephemeral**: Assessment results exist only in browser session memory
- **Commercial data protection**: Proprietary market data is excluded from the repository
- **Synthetic data is privacy-safe**: Generated distributions contain no real investor or company information